

Final Report
On
“Fire Fighter Robot Using Arduino”

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I also extend my sincere appreciation to the laboratory assistants and technical staff of the ECE Department for their timely help, technical support, and for providing the necessary laboratory facilities required for designing, testing, and implementing the system.

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Abstract

Fire accidents are one of the most critical hazards in industrial and domestic environments, often occurring in areas that are dangerous or inaccessible for humans. This project presents a Fire Fighter Robot designed using IoT-based systems to provide a safe and efficient solution for fire detection and suppression. The robot integrates Arduino controllers, flame sensors, motor drivers, and a water pump, allowing it to autonomously detect nearby fires and extinguish them while providing real-time alerts via LEDs and a buzzer.

To enhance operational flexibility, the system employs a dual ESP32 architecture. One ESP32-S module handles Bluetooth communication, enabling remote manual control of the robot, while a separate ESP32-CAM module provides live video streaming over Wi-Fi for long-distance fire monitoring. This design allows simultaneous remote supervision and automated fire suppression, demonstrating a practical application of IoT in robotics for safety-critical environments. The combination of manual and automatic modes ensures timely response, minimizes human risk, and showcases the potential of IoT-enabled robotic systems in real-world fire safety applications.

INTRODUCTION:

Fire accidents pose a serious threat to human life and property, often occurring in places where quick access or manual intervention is difficult. Traditional firefighting methods expose humans to hazardous conditions, making it essential to develop intelligent robotic systems that can detect and suppress fires autonomously. To address this issue, our project introduces a Fire Fighter Robot that combines robotics, IoT-based systems, and sensor connectivity to create an efficient, remotely controllable firefighting unit. The robot uses an Arduino-based control system integrated with flame sensors, motor drivers, and a water pump to detect and extinguish fire in a targeted area.

system's uniqueness lies in its dual microcontroller and dual ESP32 design, where one ESP32-S module handles Bluetooth communication for remote manual control, and a separate ESP32-CAM module provides Wi-Fi-based live video streaming for long-distance monitoring and visual fire detection. This separation overcomes the limitation of using a single ESP32 for both Bluetooth and Wi-Fi simultaneously. The robot can be manually driven toward a detected fire using a Bluetooth mobile app and then automatically extinguish it once flame sensors confirm its presence. This combination of manual and automatic operation demonstrates a practical IoT-based approach to enhancing safety and minimizing fire-related risks in industrial and domestic environments.

COMPONENT USED:

Component	Quantity	Function
Arduino UNO / Nano	2	Main controllers for motor & sensor logic
ESP32-S	1	Bluetooth module for manual remote control
ESP32-CAM (AI-Thinker)	1	Live video streaming and flame detection
L298N Motor Driver	1	Drives two DC motors
DC Motors + Chassis	2	Robot movement
Flame Sensors	3	Short-range fire detection (left, center, right)
Water Pump + Relay Module	1	Fire suppression unit
Buzzer & LEDs	1 each	Alerts and status indication
Battery Pack	1	Power supply for motors and control boards
Bluetooth Mobile App	1	Manual control interface

METHODOLOGY:

The methodology of this project is divided into hardware setup, software development, and integration of manual and automatic modes:

Hardware Setup

- Arduino 1: Receives Bluetooth commands from ESP32-S and controls the motors via L298N motor driver.
- Arduino 2: Connected to ESP32-CAM for long-range fire monitoring and live streaming; reads flame sensors and controls buzzer, LEDs, and water pump.
- Flame sensors (left, center, right) confirm nearby fire.
- Water pump is activated via relay to extinguish the detected flame.

Software Development

- Arduino 1 Code: Receives Bluetooth commands and moves the robot accordingly.
- Arduino 2 Code: Monitors flame sensors, triggers alarms, controls water pump, and streams live video through ESP32-CAM.
- ESP32-S Code: Communicates with Arduino #1 via Bluetooth terminal app to send movement commands.
- ESP32-CAM Code: Provides live video over Wi-Fi.

Integration & Operation

- Manual Mode: Arduino 1 moves robot via Bluetooth commands; operator sees live feed from ESP32-CAM.
- Automatic Fire Response: Arduino 2 monitors flame sensors; if fire detected, it activates alarm and pump.
- Dual Arduino setup allows simultaneous manual navigation and autonomous fire detection, overcoming ESP32 hardware limitations.

Testing

- Each subsystem (manual control, motor driver, flame sensors, pump, video streaming) was tested individually.
- The full system was tested in combination to verify coordinated response: approach fire, detect flame, stop, and extinguish.

Working Principle

Manual Mode:

- The user connects a smartphone to the **ESP32-S Bluetooth module**.
- Commands (F, B, L, R, S, P) are sent via a Bluetooth terminal app to control robot movement.
- Arduino #1 interprets these commands and drives the motors accordingly through the L298N motor driver.
- The **water pump** can also be triggered manually if needed.

Automatic Mode:

- The **ESP32-CAM** streams live video for monitoring distant fire.
- The operator manually navigates the robot toward the fire using Bluetooth control.
- When the robot is near the flame, **short-range flame sensors** detect the heat or light and send signals to Arduino #2.
- The robot automatically stops, activates the **water pump** for 3–5 seconds, and extinguishes the fire.
- The **buzzer and red LED** indicate active fire detection, while the **green LED** shows safe operation.

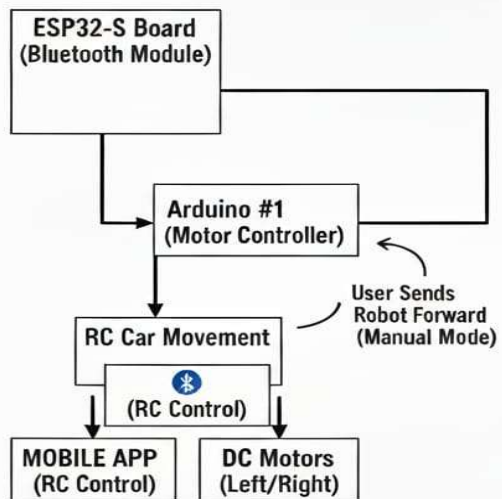
Two-Controller Coordination:

- **ESP32-S** handles Bluetooth communication (since Wi-Fi and Bluetooth cannot function together on a single ESP32-CAM).
- **ESP32-CAM** handles video streaming and long-distance fire monitoring.
- Both Arduinos work in coordination to manage movement, detection, and suppression smoothly, ensuring accurate timing and reliable performance.

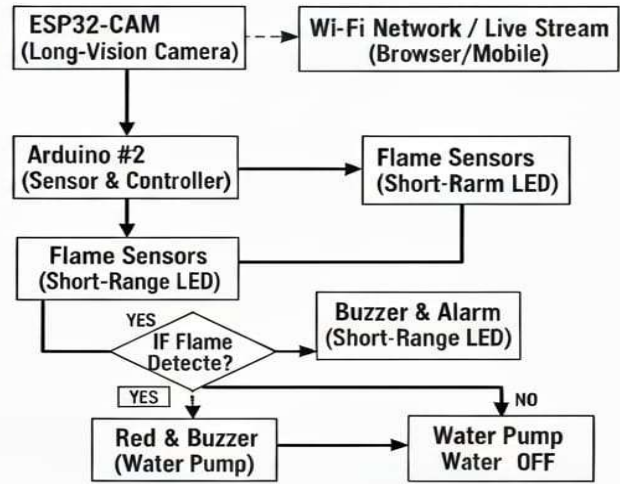
BLOCK DIAGRAM:

FIRE FIGHTING ROBOT SYSTEM

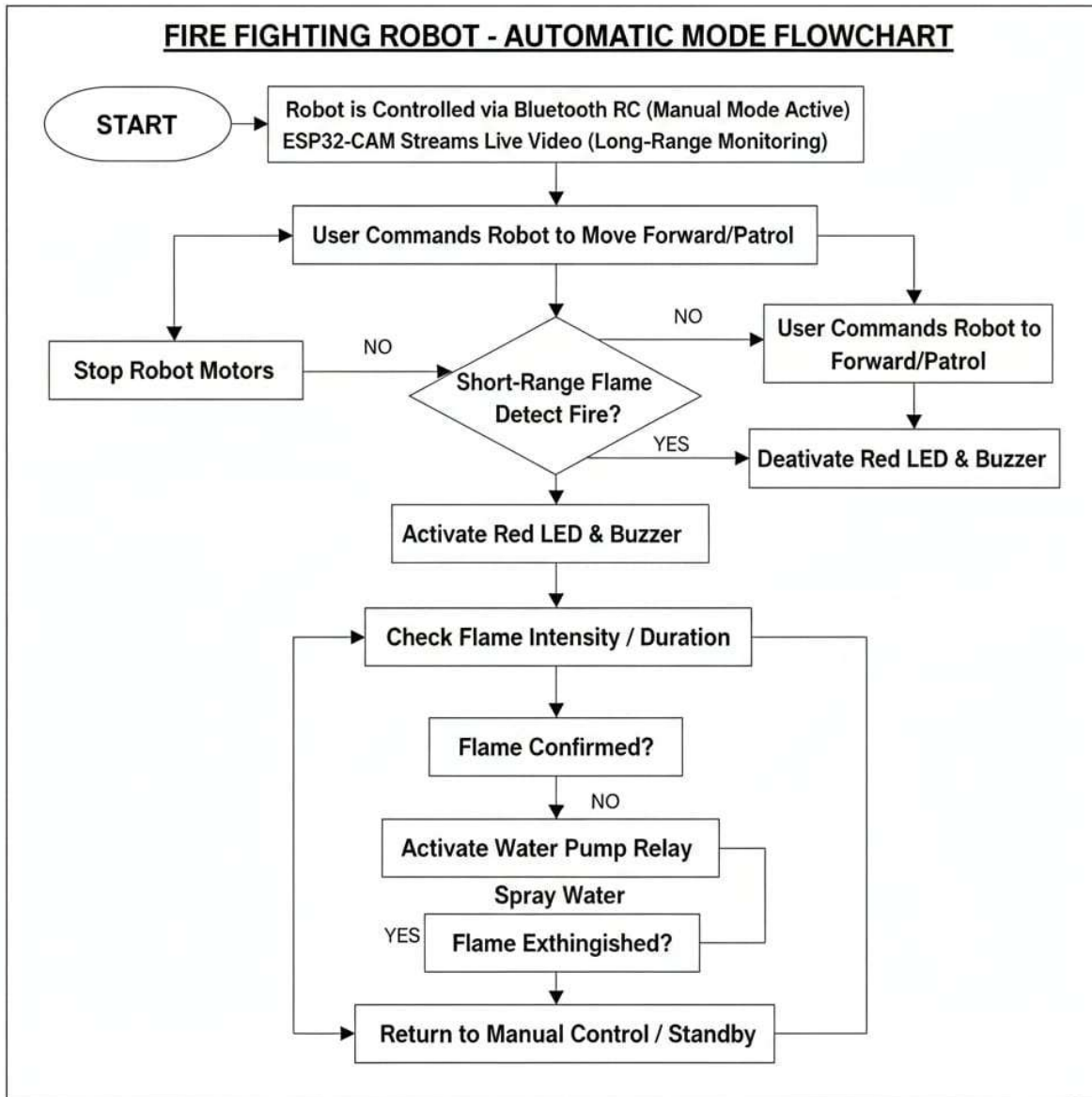
BLUETOOTH RC CONTROL



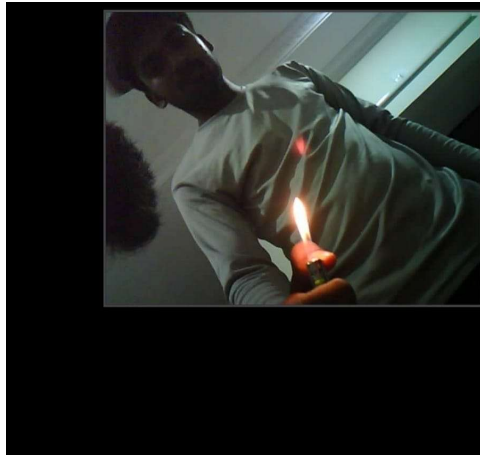
AUTOMATIC FIRE DETECTION & SUPPRESSION



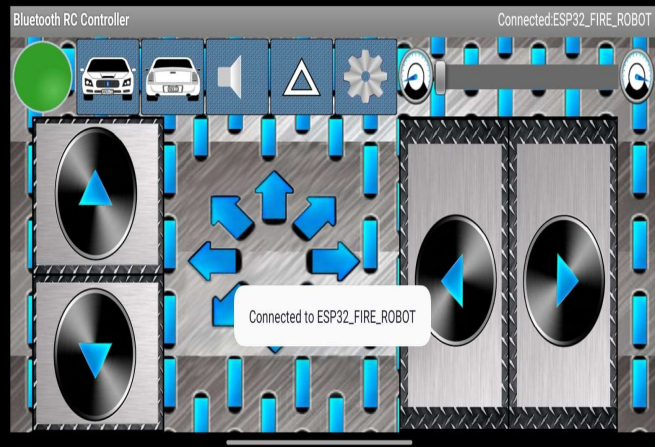
FLOWCHART:



RESULTS:



Camera Live streaming

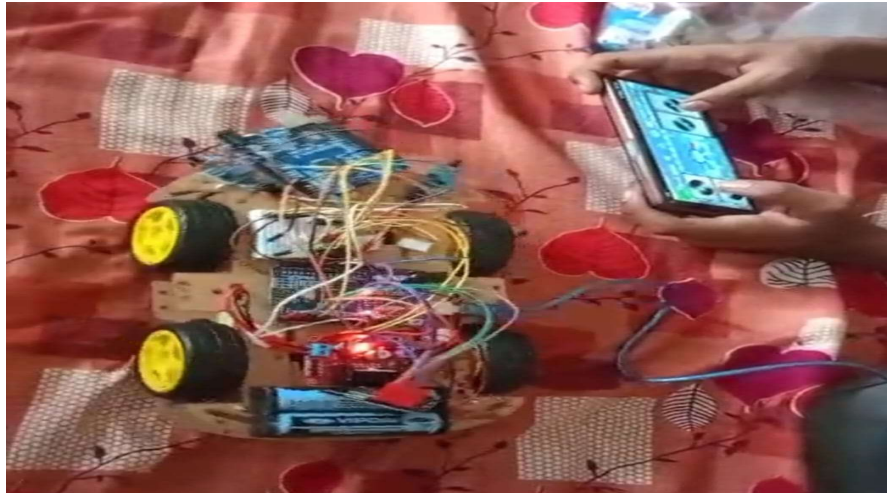


Bluetooth RC controller for Robot movement
manually



Overall robot body

when flame detected water pouring



controlling Robot through mobile phone

- The robot successfully responds to Bluetooth commands for manual navigation.
- ESP32-CAM provides stable live streaming of the surrounding environment for early fire detection.
- Flame sensors detect fire within close range and trigger the alarm system accurately.
- Automatic activation of the water pump extinguishes the flame in approximately **3–5 seconds**.
- The LED indicators clearly represent fire/safe states.
- The two-controller configuration allowed **simultaneous Bluetooth and Wi-Fi functionality** without interference.

Discussion of Results:

- The IoT-based Fire Fighter Robot performed efficiently in both **manual** and **automatic** modes. The separation of tasks between two microcontrollers (Arduino 1 & 2) ensured reliability and smooth operation. The integration of **ESP32-S** for Bluetooth and **ESP32-CAM** for Wi-Fi streaming eliminated hardware communication conflicts.
- The robot demonstrated accurate flame detection and quick response with the water pump, minimizing extinguishing delay. Real-time video feedback helped in remote monitoring, while manual override ensured human control when necessary.

- This project successfully showcases the combination of **IoT communication, robotics, and automation** to create a cost-effective, scalable solution for fire safety. Future improvements could include **temperature sensors, GPS tracking, and cloud connectivity** for enhanced automation and reporting.

Advantages

- Dual-mode operation (manual & automatic).
- Long-distance monitoring via live camera feed.
- Safe suppression of fire without human intervention.
- Low-cost implementation using Arduino and ESP32.

FUTURE WORK:

1. AI-Enhanced Fire Detection:

The system can be upgraded with **AI and image processing algorithms** on the ESP32-CAM to automatically identify fire, smoke, or heat intensity from live video, improving accuracy and response time.

2. IoT Cloud Connectivity:

Integration with **IoT cloud platforms** such as ThingSpeak or Blynk can enable real-time monitoring, remote control, and data analysis through mobile or web applications.

3. GPS and Alert System:

Adding **GPS and GSM modules** can help track the robot's location and send instant fire alerts or status updates to users and emergency services.

4. Autonomous Navigation:

Incorporating **ultrasonic or IR sensors** with path-planning algorithms will allow the robot to automatically navigate around obstacles and reach the fire source without manual control.

5. Industrial and Smart Building Use:

The robot can be adapted for **industrial zones, laboratories, and smart buildings**, where continuous monitoring and rapid fire response are critical for safety automation.

CONCLUSION:

The proposed IoT-based Fire Fighter Robot successfully integrates multiple technologies—Arduino microcontrollers, ESP32 modules, flame sensors, and a water pump system—to create an efficient and autonomous fire detection and suppression mechanism. By dividing the control system into two dedicated units, the project effectively overcomes the limitations of using a single microcontroller for both Bluetooth and Wi-Fi operations.

The robot demonstrated reliable performance in both manual and automatic modes. Manual control through Bluetooth (ESP32-S) allowed precise movement and directional control, while automatic operation using flame sensors and ESP32-CAM video streaming enabled rapid flame detection and extinguishing. The combination of live monitoring and intelligent response mechanisms ensures both user safety and effective hazard mitigation.

This project not only highlights the potential of IoT in real-time hazard management but also provides a foundation for further enhancements such as temperature sensing, GPS-based navigation, and cloud data integration for smart emergency systems. Overall, the system proves to be a cost-effective, scalable, and innovative solution for real-world fire safety applications in homes, industries, and laboratories.

Mini Project1 Report – Fire Detection System using Arduino

Group Members:

- 1) S. Mahaboob Hussain -AP23110020132
- 2) B. Kotesb -AP23110020141

1) Introduction

Fire accidents are one of the most common hazards in industries, homes, and public places. Early detection of fire can save lives and property by providing timely alerts. Traditional fire detection systems are often costly and complex. In this mini project, we present a low-cost fire detection system using Arduino Uno and flame sensors. The system detects the presence of flame and provides visual and audio alarms using LEDs and a buzzer.

2) Methodology

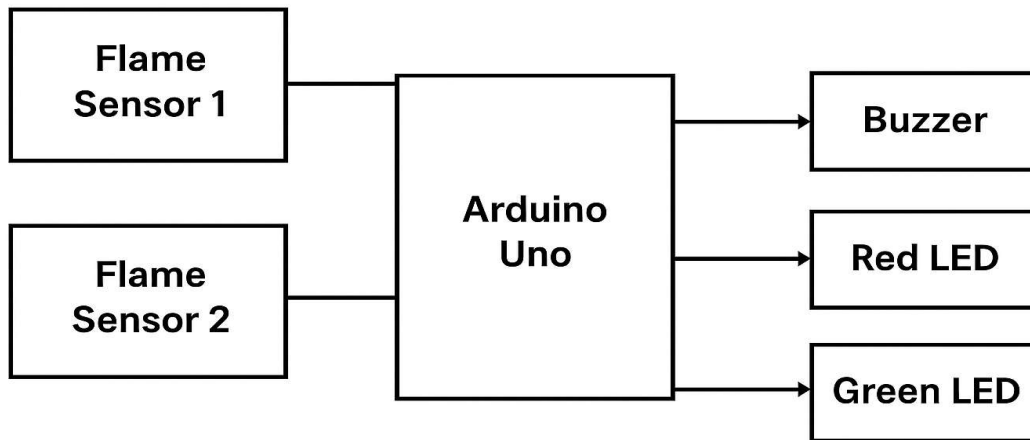
The project uses Arduino Uno as the main controller. Two flame sensors are placed in different directions to detect fire presence. When any sensor detects fire, it sends a signal to Arduino. The Arduino then immediately activates a buzzer and a red LED to indicate danger. When there is no fire, a green LED remains ON to indicate safe conditions.

Steps followed:

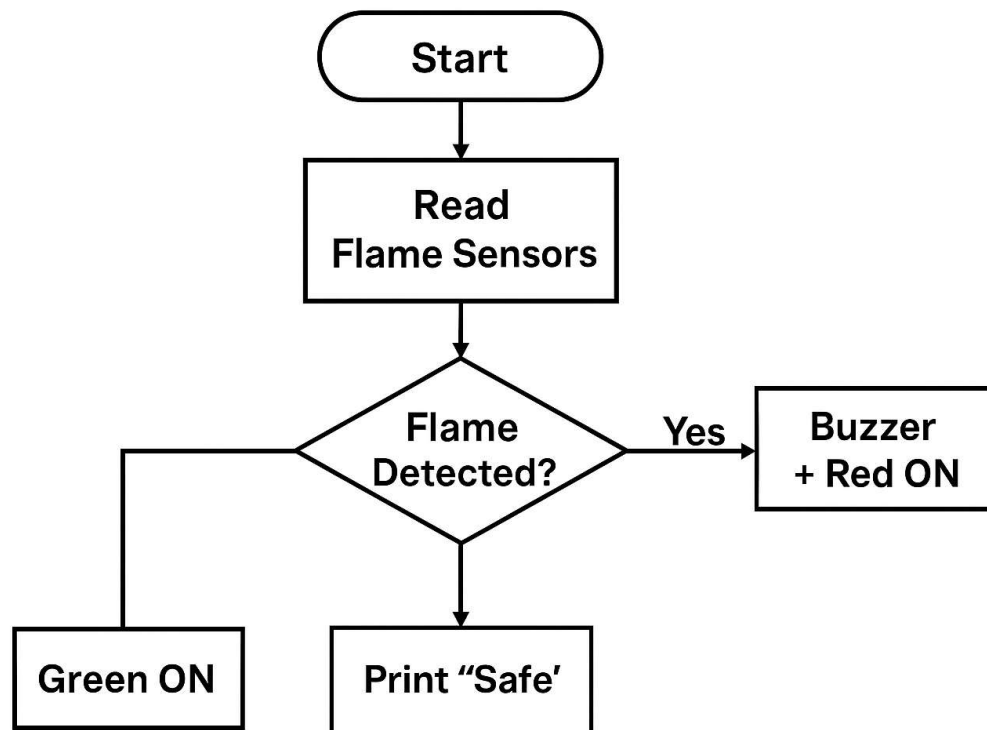
- **Step 1:** Connect two flame sensors to Arduino input pins.
- **Step 2:** Connect a buzzer and two LEDs (Red for fire alert, Green for safe).
- **Step 3:** Write an Arduino program to continuously read flame sensor values.
- **Step 4:** If flame is detected, activate buzzer + red LED, else keep green LED
- **Step 5:** Display system status (Flame detected / Safe) on Arduino IDE Serial Monitor

3)Block Diagram

BLOCK DIAGRAM



Flow Chart



4) Discussion of Results

The project successfully demonstrates how low-cost components can be used to build a fire detection system. The system works in real-time and responds quickly when a flame is detected. The use of two sensors increases coverage area, ensuring that fire from either direction is detected. The buzzer provides an immediate audio alert, while LEDs give clear visual status.

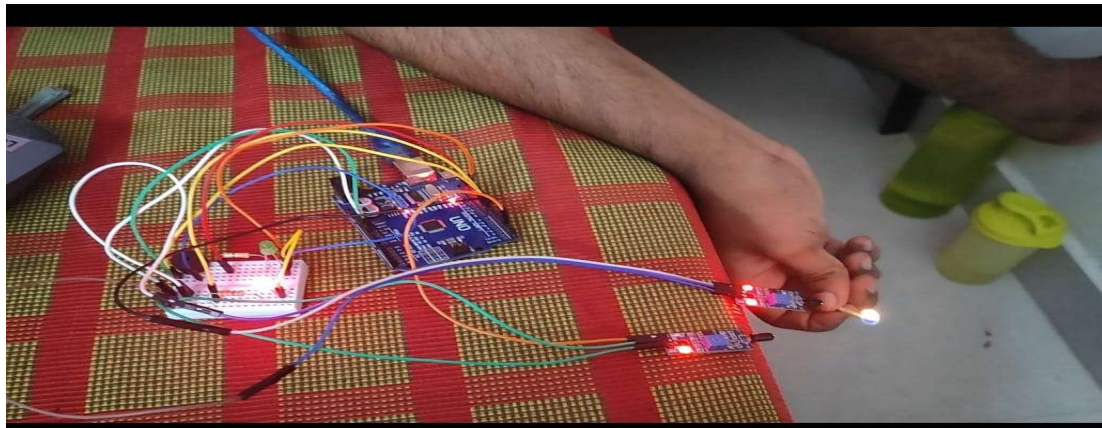
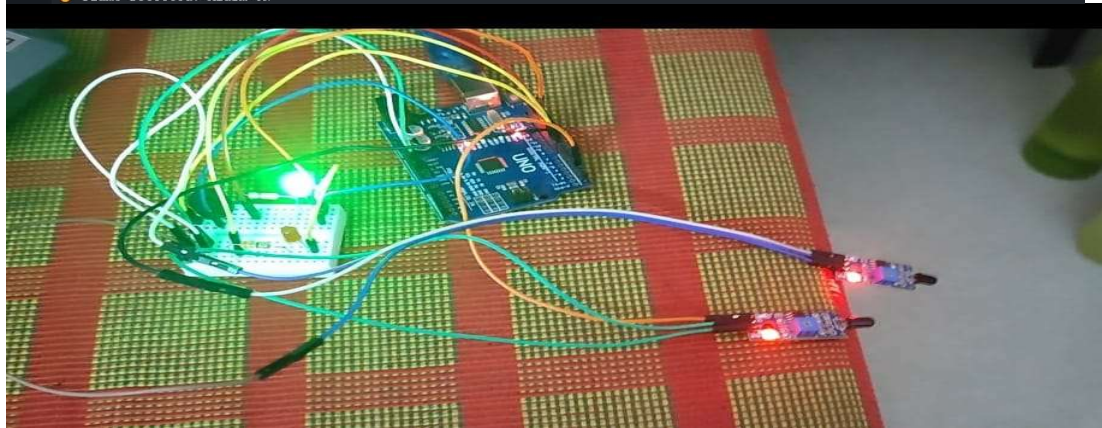
5)Results

```
File Edit Sketch Tools Help
Arduino Uno
sketch_sep3a.ino
3
4 int flame1 = 2; // Flame Sensor 1 DO pin
5 int flame2 = 3; // Flame Sensor 2 DO pin
6 int buzzer = 8; // Buzzer pin
7 int redLed = 10; // Red LED pin
8 int greenLed = 11; // Green LED pin
9
10 void setup() {
11   pinMode(flame1, INPUT);
12   pinMode(flame2, INPUT);
13   pinMode(buzzer, OUTPUT);
14   pinMode(redLed, OUTPUT);
15   pinMode(greenLed, OUTPUT);
16
17   Serial.begin(9600); // Start Serial Monitor
18   Serial.println("🔥 Fire Detection System Started");
19 }
20
21 void loop() {
22   int f1 = digitalRead(flame1); // Read sensor 1
23   int f2 = digitalRead(flame2); // Read sensor 2
24 }
```

Output Serial Monitor x

Message (Enter to send message to 'Arduino Uno' on 'COM5')

🔥 Flame Detected! Alarm ON
🔥 Flame Detected! Alarm ON



5) Extension of Work

In the next stage of development, the system can be extended into a fire-fighting robot. Instead of just detecting fire and raising an alarm, the robot can be programmed to move in the

direction of the detected flame. This enhances the system by providing active response rather than just passive alerting.

Long-Range Fire Detection (Vision System)

- The camera + onboard ML / image processing is used to detect and localize flames at a longer distance.
- The vision processor (Raspberry Pi or ESP32-CAM) analyzes the camera feed and identifies flame direction (left, right, or centre).
- Based on this analysis, it sends simple navigation commands (LEFT / RIGHT / FORWARD) to the Arduino or motor driver.

Short-Range Fire Confirmation (Flame Sensors)

- Once the robot approaches the flame, the flame sensors take over.
- Sensor-based navigation ensures accurate movement:
 - Left sensor ON → Move Left
 - Centre sensor ON → Move Forward
 - Right sensor ON → Move Right
 - Multiple sensors ON → Move Forward (towards source)

Mini Project 2 Report -- Long-Range Vision Fire Detection & Navigation

Group members:

- 1) S. Mahaboob hussain -AP23110020132
- 2) B. Kotesesh -AP23110020141

01) Introduction :

Fires are one of the most damaging hazards in homes, industries, and public spaces. Traditional fire detection systems are often fixed in place and only alert users. This project presents a low-cost, mobile fire-fighting robot using ESP32-CAM (for vision-based detection) and Arduino with flame sensors (for close-range confirmation and fire extinguishing). The system autonomously detects, navigates to, and extinguishes fire, demonstrating how embedded systems, image processing, and sensor integration can increase safety and response efficiency.

02) Methodology :

Long-Range Fire Detection: An ESP32-CAM module scans camera images to detect and locate flames. When fire is spotted, it sends navigation commands indicating the fire's direction (left, center, right) through serial to the Arduino.

Short-Range Confirmation & Action:

The Arduino receives direction commands and moves the robot accordingly using an L298N motor driver. When the robot nears the fire, dedicated flame sensors check for precise flame location. If a fire is confirmed, the robot stops and activates a water pump to spray water and put out the fire.

Steps followed:

Step 1: Connect ESP32-CAM for RGB image analysis and serial communication.

Step 2: Interface Arduino with L298N motor driver and three flame sensors.

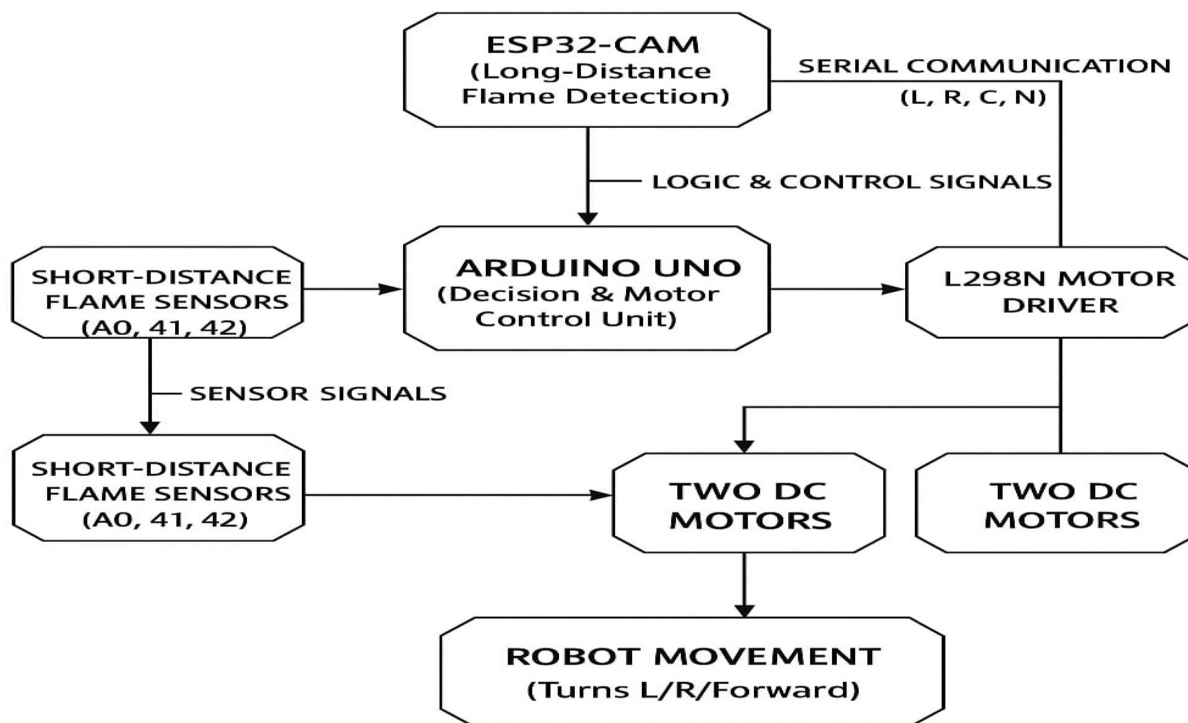
Step 3: Write ESP32-CAM code to analyze images and send serial commands ('L', 'C', 'R', 'N') based on fire direction.

Step 4: Write Arduino code to:

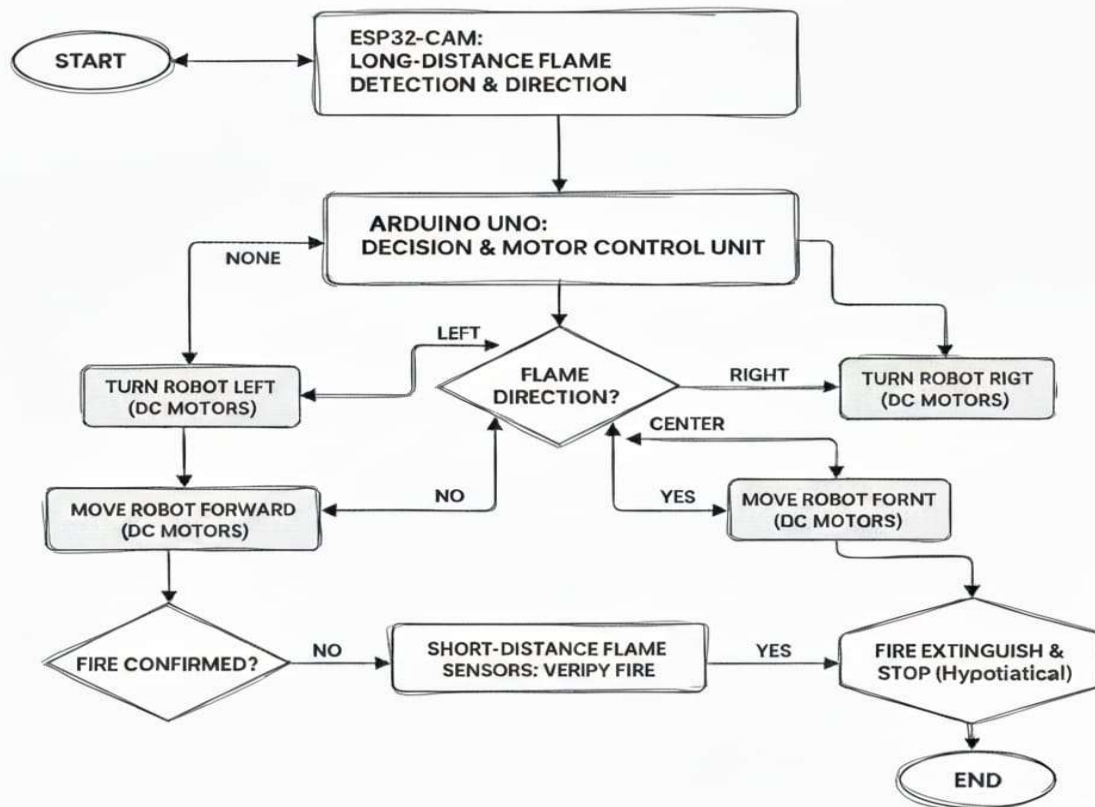
- Drive the robot's motors (move left, forward, right, or stop according to serial commands).
- Monitor flame sensors for close-range fire detection.

Step 5: Assemble, wire, and test the complete system to ensure reliable detection, navigation, and water-based extinguishing.

03) Block diagram :



04) Flowchart :

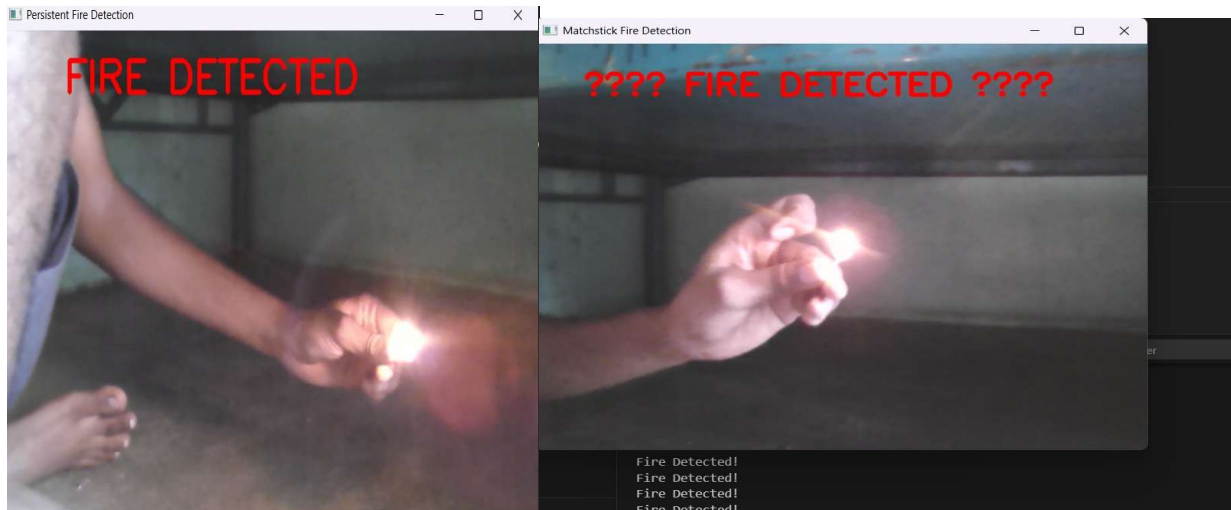


05) Results And Discussion :

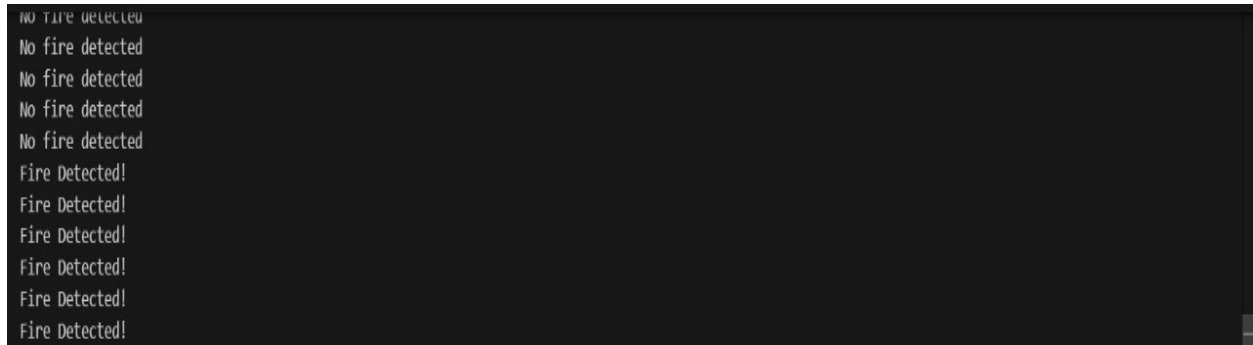
The multi-stage detection system effectively combines vision and sensor data to detect and extinguish fire autonomously. The camera-based long-range detection guides the robot toward fire zones, while short-range flame sensors provide safety confirmation. Water is used to extinguish the detected fires effectively. This layered approach reduces false positives and ensures targeted fire extinguishing. In this project, we focused on using the ESP32-CAM for long-distance fire detection and robot navigation toward the fire location. The water-based extinguishing system can be integrated later for actual fire suppression

Results :

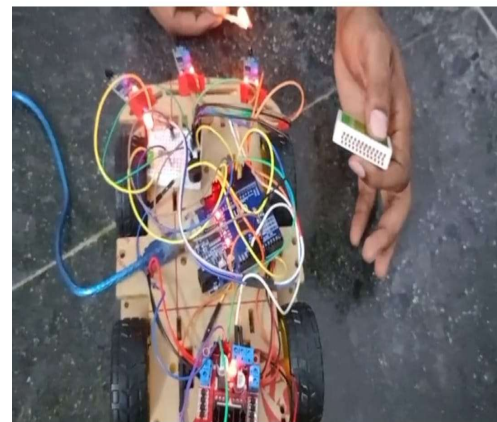
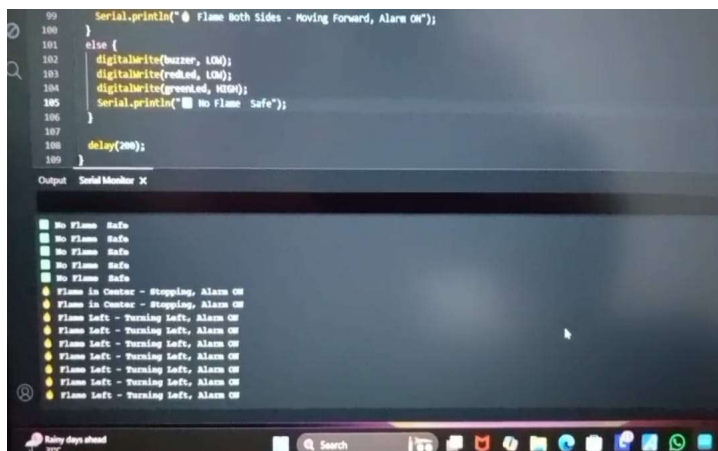
Long range fire detection using ESP32 cam



Before fire is not detection



Short range fire detection and movement



06) Extension of work:

1.Integration of Water Pump System:

- Add a DC water pump controlled by the Arduino relay module.
- When nearby flame sensors confirm a fire, the relay will automatically activate the water pump to spray water and extinguish the flame.
- This will transform the robot from a detection-only system to a fully functional firefighting robot.

2.Enhanced Automation:

- Enable the robot to autonomously detect, approach, and suppress fires without human control.
- Ensure smooth communication between the ESP32-CAM and Arduino Uno for synchronized operation.

3.Improved Detection Accuracy:

- Enhance image-processing algorithms for better long-distance flame recognition.
- Fine-tune detection parameters to minimize false alarms caused by ambient light or reflections.

4.Upgraded Hardware Control:

- Optimize Arduino control logic for managing motors and extinguishing relays simultaneously.
- Maintain stable and isolated power supply for both motor and camera circuits to prevent resets.

5.System Enhancement for Real-World Application:

- Integrate safety checks, feedback loops, and thermal shielding for field deployment.
- Transform the system into a fully autonomous firefighting robot capable of real-time detection and suppression.

07) conclusion:

The developed fire detection and movement system successfully demonstrates a multi-stage approach that combines both vision-based and sensor-based technologies. The ESP32-CAM enables long-distance fire detection and guides the robot toward the fire zone, while flame sensors provide accurate short-range verification of the fire's presence. This layered detection method minimizes false positives and ensures reliable navigation toward actual fire sources. The system effectively achieves autonomous movement toward detected flames, proving its capability as an intelligent fire-tracking robot. Although the current prototype focuses on fire detection and navigation, future enhancements will include the integration of a water pump system to automatically extinguish fires when detected, thereby transforming it into a fully autonomous firefighting robot.